

Edge-Optimized Multi-Task Deep Learning for Joint Classification and Segmentation of Lung Ultrasound Images.

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Abstract— Lung ultrasound (LUS) has emerged as a critical imaging modality for respiratory disease diagnosis in low-resource settings, yet its interpretation requires specialized expertise that is severely limited in Uganda, which has only 0.16 radiologists per 100,000 population. This study develops an edge-optimized multi-task deep learning framework for simultaneous classification and segmentation of LUS images to assist in respiratory disease diagnosis. Using publicly available datasets of 1,062 LUS images from Mulago and Kiruddu National Referral Hospitals (classified into COVID-19, other lung diseases, and healthy) and 401 images with B-line annotations, we propose an EfficientNet encoder paired with a SegFormer decoder for multi-task learning. The model achieves 77.1% classification accuracy and 0.68 Dice coefficient for B-line segmentation. To ensure clinical utility in resource-constrained settings, the model was exported to TorchScript and deployed on Android smartphones via PyTorch Mobile, achieving very good inference time on a Samsung Galaxy S9 and A17, operating entirely offline without internet dependency. Explainable AI techniques including Grad-CAM for classification and attention maps for segmentation, provide visual verification that the model focuses on clinically relevant B-line regions. This work demonstrates that edge-deployed multi-task deep learning can democratize access to expert-level LUS interpretation in low-resource healthcare settings, directly addressing the radiologist shortage in Uganda while maintaining patient data privacy through offline operation. The deployed application is currently aimed for clinical pilot testing at Mulago National Referral Hospital.

Keywords— *Edge Computing (TinyML), Explainable AI (Grad-CAM), Lung Ultrasound (LUS), Multi-task Learning, Resource-Constrained Diagnostics, Semantic Segmentation.*

I. INTRODUCTION

Lung ultrasound (LUS) is a portable, low-cost and radiation-free imaging technique suitable for diagnosing respiratory diseases in resource-constrained settings such as Uganda, where there is a severe shortage of radiologists, especially in rural areas. Ultrasound artifacts such as B-lines, pleural abnormalities and consolidations are clinically associated with diseases including pneumonia, pulmonary edema and COVID-19 thus creating a need for automated AI-based diagnostic support.

This research applies multi-task deep learning to simultaneously perform disease classification and pixel-level

segmentation of lung ultrasound abnormalities. The framework uses an EfficientNetB3 encoder, SegFormer decoder, transfer learning and attention mechanisms to improve feature learning, interpretability and segmentation precision. Grad-CAM and attention maps are integrated to generate clinically meaningful visual explanations that improve transparency and trust.

The study addresses key limitations in existing approaches, including classification-only models, noisy bounding-box annotations and lack of optimization for low-resource deployment. A major computational challenge is that noisy ultrasound data, speckle noise and class imbalance cause classification gradients to dominate segmentation gradients hence leading models to learn spurious correlations instead of clinically relevant features.

To overcome these issues, the proposed framework is optimized for Edge/TinyML deployment using techniques suitable for smartphones and low-power devices. The final model is deployed offline on Android devices using PyTorch Mobile thus enabling practical AI-assisted lung ultrasound diagnosis in rural Ugandan healthcare settings.

II. RELATED WORK

Early multi-task learning studies demonstrated that unified architectures can jointly perform classification and segmentation using shared feature representations, improving generalization and diagnostic performance[1]. Subsequent optimization research introduced task-adaptive methods such as the EVE optimizer[2] to reduce gradient imbalance and improve stability across multiple tasks.

Interpretability-focused models such as MIB-Net[3] highlighted the importance of explainable AI by generating contribution maps that improve transparency and clinical trust in medical imaging systems. For edge deployment, lightweight architectures such as USEANet[4] showed that efficient segmentation models with low computational complexity can achieve strong performance on portable and low-resource devices.

In the Ugandan context[5, 6], publicly available lung ultrasound datasets provide locally relevant data for disease

classification and B-line segmentation. However, existing approaches remain limited to single-task learning and rely on noisy bounding-box annotations. Overall, the literature reveals a critical gap in developing an interpretable, edge-optimized multi-task framework that simultaneously performs classification and precise segmentation for lung ultrasound analysis in resource-constrained settings.

III. PROBLEM STATEMENT

A. Clinical motivation

Uganda faces a burden of high incidences of respiratory diseases and also limited diagnostic capacities in many areas.

Disease burden: Lower respiratory infections are among the top causes for mortality nationally with pneumonia alone accounting for approximately 15% of under-five deaths .

Workforce shortage: Uganda has a small number of radiologists compared to the nation population. This shortage is most acute in rural areas where 75% of the population resides.

Diagnostic gap: Many health centers lack access to computed tomography (CT) and even basic X-ray services forcing clinicians to rely on clinical examination alone.

B. Computational problem

While the clinical imperative is clear, the underlying Deep Learning computational problem lies in the optimization instability and representation entanglement inherent in joint classification and segmentation of highly noisy ultrasound data.

Lung ultrasound images are characterized by severe speckle noise, acoustic attenuation and class imbalance where artifacts like consolidations occupy a minute fraction of the pixel space.

When training standard CNNs on this data, models rapidly overfit to background noise or exhibit gradient dominance where the larger magnitude gradients of the Binary Cross-Entropy (classification) loss overpower the localized Dice loss (segmentation).

This occurs during the backpropagation of shared encoder weights in multi-task setups because standard architectures lack the capacity to dynamically route attention to sparse, clinically relevant pixels while ignoring acoustic artifacts. Consequently, without explicit mechanisms to force the network to focus on causal morphological features (like pleural line irregularities), the model relies on spurious correlations leading to catastrophic failure in Out-of-Distribution (OOD) clinical deployment.

An additional challenge arises in the context of edge optimization and deployment constraints, particularly in low-resource healthcare environments. Lung ultrasound is often

performed using portable or handheld devices with limited computational capacity, memory and power availability. However, many state-of-the-art multi-task architectures especially those incorporating transformers, attention modules or multi-branch networks introduce significant computational overhead, making them impractical for real-time inference on edge devices. Moreover, optimization techniques designed to balance multi-task learning such as gradient modulation or adversarial regularization can further increase training and inference complexity. This creates a critical trade-off between model performance and deployability, where highly accurate models may be too resource-intensive for practical use. Consequently, there is a need for lightweight, computationally efficient architectures and optimization strategies that can maintain robustness and accuracy while operating within the strict constraints of edge environments, ensuring real-time, reliable clinical decision support in settings such as rural Ugandan healthcare systems.

IV. PROPOSED METHODOLOGY

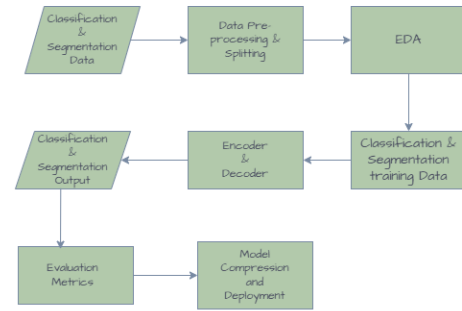


Figure 1. Illustrates the proposed methodology for the development of the Multi-Task system.

The study utilizes two Ugandan lung ultrasound datasets: classification dataset containing 1,062 images labeled into COVID-19, healthy and other disease classes and a segmentation dataset containing 401 images annotated for B-lines using polygon-based masks. Exploratory Data Analysis (EDA) is performed to examine class distributions, dataset splits and B-line localization patterns through pie charts, histograms and heatmaps.

For preprocessing, all images and masks are resized to 224×224 pixels and normalized using ImageNet statistics. Data augmentation including rotation, flipping, brightness/contrast adjustment and Gaussian noise injection is applied only to the training set to improve generalization. YOLO polygon annotations are converted into binary segmentation masks using rasterization techniques while preserving boundary information through nearest-neighbor interpolation.

To ensure unbiased evaluation and prevent data leakage, strict patient-held-out splitting is enforced such that images

from the same patient do not appear across training, validation and test sets. Class imbalance is addressed using inverse-frequency class weighting, stratified sampling and balanced composite loss functions. All images are retained to preserve real-world clinical variability and noise characteristics found in Ugandan healthcare settings.

The proposed framework employs a shared encoder-decoder multi-task architecture using an EfficientNet-based encoder and SegFormer-style decoder with attention mechanisms to capture both global and local ultrasound features while suppressing acoustic artifacts. The model simultaneously performs disease classification and pixel-level B-line segmentation through separate classification and segmentation heads.

Training is performed using joint optimization with a weighted combination of classification loss and region-based segmentation loss to reduce gradient interference and improve balanced feature learning. Grad-CAM and attention maps are incorporated to provide interpretable visual explanations of model predictions.

Model performance is evaluated using classification metrics including accuracy, precision, recall, F1-score and AUC, alongside segmentation metrics such as Dice coefficient and IoU. Finally, the trained model is optimized through compression and conversion to TorchScript/edge-compatible formats for efficient offline deployment on Android smartphones and low-resource edge devices using PyTorch Mobile.

V. EVALUATION METRICS

TABLE 1. ILLUSTRATES THE ACCURACY AND DICE SCORE OF THE TRAINED MODELS

Model Architecture	Recall	Accuracy	Dice Score	F1 Score
ResNet50 + Attention	46.5%	100%	0.433	100%
EfficientNet-B3 + SegFormer	71%	77.1%	0.68	70%
ConvNeXt + DeepLabV3+	21.89%	39.01%	0.000	21.89%
MobileNetV3 + FPN	21.89%	39.01%	0.000	21.89%
MobileNetV3 + Attention U-Net	21.99%	21.99%	0.000	7.93%

EfficientNet-B0 + Attention	21.99%	22%	0.000	7.93%
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TABLE 2. ILLUSTRATES THE EXPLAINABLE AI OF TRAINED MODELS

Model Architecture	Explainable AI Performance
ResNet50 + Attention	Grad-CAM aligned the radiologist-identified pathology with hierarchical attention, demonstrating good XAI performance.
EfficientNet-B3 + SegFormer	Grad-CAM highly aligned with radiologist-identified pathology achieving strong segmentation performance and proving most suitable for deployment.
ConvNeXt + DeepLabV3+	Grad-CAM outputs were random and ASPP visualizations were uninformative, resulting in complete interpretability failure.
MobileNetV3 + FPN	Poor Grad-CAM results with no meaningful feature patterns, making the model fast but uninterpretable.
MobileNetV3 + Attention U-Net	Attention maps collapsed into uniform activation and Grad-CAM outputs were random.
EfficientNet-B0 + Attention U-Net	Attention collapse occurred with no spatial selectivity, leading to complete interpretability failure.

Note that the Grad-CAM performance is related to the Dice Score.

VI. MODEL COMPRESSION AND DEPLOYMENT

TABLE 3. ILLUSTRATES HOW THE COMPRESSION AND OPTIMIZATION TECHNIQUES AFFECTED OUR TRAINED MODEL.

Technique	Impact	Decision
FP16 (Half Precision)	Reduced model size by 50% (57.19MB to 28.60MB) without accuracy or Dice score loss.	Used
INT8 Quantization	Reduced segmentation quality and Dice performance to clinically unacceptable levels.	Not Used

Pruning	Decreased B-line localization precision and thus segmentation quality.	Not Used
TorchScript Export	Improved mobile inference speed by approximately 2–3× for efficient deployment.	Used
ONNX Export	Caused instability and conversion failures due to EfficientNet compatibility issues.	Not Used
Dynamic Quantization	Provided minimal compression benefits compared to FP16. (57.19MB to 56.87MB)	Not Used

The model was then deployed on streamlit and can be accessed through the link: <https://lung-ultrasound.streamlit.app/>.

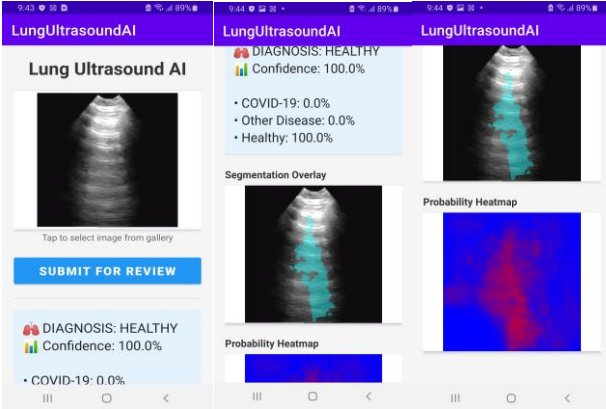


Figure 2. Illustrates the interface of the model as accessible from Samsung S9.

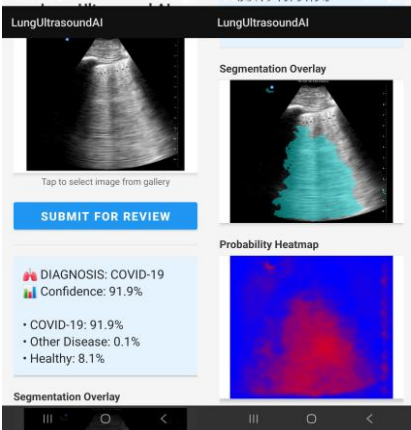


Figure 3. Illustrates the interface of the model on Samsung A17.

After evaluation of the compression and optimization methods, the best approaches were chosen. Following through, the model was compressed with FP16(Half Precision), converted to TorchScript format making it deployment ready and then Android Studio was used to put and use the model within a mobile app on edge devices.

VII. LIMITATIONS AND FUTURE WORKS

Despite the promising performance of the proposed multi-task lung ultrasound framework, several limitations remain. One major challenge is the risk of domain shift, where variations in ultrasound devices, acquisition protocols and imaging environments may reduce model generalization across different clinical settings. In addition, the use of bounding-box pretraining for B-line localization introduces label noise because bounding regions may contain extraneous healthy tissue thus potentially affecting segmentation precision and optimization stability.

Future work will focus on improving generalization through domain adaptation techniques that enable the model to perform consistently across different ultrasound devices and healthcare environments. Active learning strategies will also be explored to expand the dataset efficiently while minimizing annotation cost and radiologist workload. Increasing the availability of high-quality pixel-level segmentation masks will further improve boundary localization and segmentation accuracy.

Additional future directions include integration with electronic medical record (EMR) systems to support longitudinal patient monitoring and clinical workflow integration. Further optimization for real-world deployment in low-resource settings will also be pursued, including improved mobile inference performance and deployment through mobile application platforms such as the Google Play Store and Apple App Store.

VIII. CONCLUSION

This research demonstrates that strict data handling, balanced optimization and causal ablation studies can produce an interpretable and computationally efficient multi-task lung ultrasound framework for low-resource healthcare settings. The proposed system successfully combines classification and B-line segmentation while maintaining explainability through Grad-CAM and attention mechanisms. Among the evaluated models, MEfficientNetB3 + Segformer achieved the best balance between segmentation performance, interpretability and deployment efficiency, enabling offline smartphone-based inference for rural Ugandan clinics. Overall, the study highlights the potential of edge-optimized AI systems to support lung ultrasound diagnosis in resource-constrained environments.

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